[i]AI-Driven Sustainability Intelligence - AI-Driven Textile Waste Management System

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Project Proposal Report

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Particularly in developing nations like Sri Lanka, the growing volume of textile waste produced by the fashion and manufacturing industries has become a serious environmental concern. The primary methods used in traditional textile waste management are landfilling and burning, which pollute the environment and destroy important reusable materials. Through enhanced visibility, traceability, and data-driven decision making, this study presents an AI-enabled digital marketplace platform intended to promote the effective reuse and recycling of textile waste.

Fabric categorization, trash value, environmental impact prediction, traceability systems, and an online marketplace for trading textile waste are just a few of the sophisticated modules that are integrated into the suggested system. The marketplace platform and analytics component of this system concentrate on offering a centralized digital environment where buyers, recyclers, and producers of textile waste can interact and trade materials. Additionally, a predictive waste analytics dashboard is created to estimate future patterns in textile waste by analysing past waste creation data.

Stakeholders may make well-informed decisions about resource allocation, waste collection, and reuse options thanks to the forecasting capacity. The technology encourages a circular economy approach to textile waste management by fusing marketplace functionality with predictive analytics. The proposed platform seeks to lessen the impact on the environment, increase the transparency of textile waste flows, and assist the textile industry in making sustainable decisions.

Acknowledgement

I want to sincerely thank my supervisor for all their help and advice during this study assignment. Their insightful counsel and support were crucial to this study's effective conclusion.

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List of abbreviations

AI	Artificial Intelligence
UN	United Nation
SDG	Sustainable Development Goal
MDPI	Multidisciplinary Digital Publishing Institute
SMEs	Small and Medium-sized Enterprises
ISO	International Organization for Standardization
API	Application Programming Interface
JSON	A lightweight data format used for data exchange between systems.

1. Introduction

This part of the study focuses on creating an **analytics system and digital marketplace platform** to manage textile waste effectively. The platform serves as a central location for the listing, discovery, and exchange of textile waste materials by recyclers, textile manufacturers, and other stakeholders. Digital platforms are increasingly recognized as effective tools for improving resource circulation and supporting circular economy practices in the textile sector. Users can upload details about the **type, quantity, and location of available textile waste** to the marketplace. In contrast to conventional disposal techniques like **landfilling or incineration**, this promotes the reuse and recycling of textile waste and increases the visibility of recyclable materials.

The solution also includes a **waste forecasting analytics dashboard** in addition to marketplace features. The dashboard estimates future trends in textile waste by analysing past waste generation data and applying **predictive analytics techniques**. Predictive models have been widely used in waste management studies to forecast waste quantities and support better planning of recycling and resource management strategies.

Additionally, the analytics component offers **visual insights through metrics and charts** that facilitate well-informed decision-making. By combining the marketplace with predictive waste analytics, the technology improves the effectiveness of textile waste management and encourages sustainable practices in the textile sector.

1.1 Background and Context

1.1.1Real-World Problem

The production of textile waste has significantly increased because of the fast expansion of the worldwide fashion and textile industries. Both post-consumer garment disposal and manufacturing operations generate significant amounts of textile waste. Due to ineffective waste management systems and inadequate data visibility, a large portion of this garbage is burned or dumped in landfills. It is frequently challenging to efficiently reuse or recycle textile waste materials because information regarding their kind, amount, and availability is not appropriately recorded or shared among stakeholders.

1.1.2 Who Is Affected and Why It Is Important

The issue of textile waste affects multiple stakeholders, including textile manufacturers, recycling companies, environmental organizations, and policymakers. Manufacturers often generate leftover fabric and material waste, but may not have efficient systems to manage or redistribute these resources. Recycling industries also face challenges due to the lack of accurate information about available textile waste materials. Without proper visibility and data analysis, opportunities for reuse and recycling are lost, leading to environmental pollution and economic inefficiencies. Therefore, improving transparency and providing analytical insights into textile waste is essential for sustainable waste management.

1.1.3 Statistical Evidence (Local and Global)

Globally, the textile industry produces approximately **million tons of textile waste annually[1]**, and this figure is expected to increase significantly in the coming years due to rising clothing consumption and fast fashion trends. Studies indicate that only a small percentage of textile waste is recycled, while the majority is disposed of in landfills. In Sri Lanka, the textile and apparel sector is one of the largest contributors to the national economy, but it also generates considerable amounts of production waste[2]. The absence of data-driven monitoring systems makes it difficult to track waste generation patterns and predict future waste quantities.

1.1.4 Analysis of Existing Solutions and Their Limitations

Several digital platforms and recycling initiatives exist to promote textile reuse and recycling. However, many of these systems primarily focus on basic listing or disposal processes and

lack advanced analytical capabilities[3]. Most existing solutions do not provide detailed visibility into textile waste materials, such as type, quantity, and availability[4]. Furthermore, they rarely include predictive models that can analyze historical waste data to forecast future waste generation. In addition, many platforms do not generate analytical reports or visual dashboards that support data-driven decision-making for stakeholders. These limitations reduce the effectiveness of current textile waste management solutions.

1.1.5 Proposed Solution and Its Key Components

To address these challenges, this research proposes the development of a **data-driven textile waste analytics platform** that improves transparency, forecasting, and decision-making. The proposed system focuses on several key components aligned with the research objectives.

First, the system will improve **textile waste visibility** by providing detailed information about textile waste materials, including their type, quantity, and availability. This will allow stakeholders to clearly understand available waste resources.

Second, a **waste forecasting model** will be developed to analyze historical textile waste data and identify patterns and trends. This predictive model will estimate future textile waste quantities, enabling better planning for reuse and recycling.

Third, the system will include **report generation and visualization features**, where analytical reports and dashboards will present insights about waste trends, predictions, and material distribution.

Finally, the platform will support **data-driven decision-making** by providing stakeholders with meaningful analytics and insights that can guide strategies for textile waste reuse, recycling, and circular economy practices.

1.1.6 Sustainable Development Goals (SDG) Alignment

Several Sustainable Development Goals (SDGs) of the UN are supported by this research. Encouraging the effective use and reuse of textile materials, it supports SDG 12 (Responsible Consumption and Production). Additionally, by lessening the environmental effects of disposing of textile waste, it promotes SDG 13 (Climate Action). Predictive analytics and digital technology integration also support SDG 9 (Industry, Innovation, and Infrastructure).

1.1.7 Research Cluster Alignment

The proposed research falls within the research clusters of **Artificial Intelligence, Data Analytics, and Sustainable Technology**. By integrating predictive modeling, analytical reporting, and digital data visibility into textile waste management, the system contributes to technological innovation that aims to improve sustainability and support circular economy initiatives.

1.2 Literature Review

The increasing environmental impact of textile waste has attracted significant research attention in recent years. Many studies focus on waste management strategies, recycling technologies, and digital systems that support sustainable textile production and circular economy practices. This section reviews relevant scholarly research related to textile waste visibility, predictive analytics, reporting systems, and data-driven decision-making.

1.2.1 Textile Waste Management and Visibility

Textile waste has become a major environmental challenge due to the rapid growth of the global fashion industry. Studies indicate that millions of tons of textile waste are generated every year, with a large portion being disposed of in landfills or incinerated due to inefficient waste management systems. Researchers emphasize that better monitoring and transparency of textile waste streams are essential for improving recycling and reuse processes[5].

However, many current waste management systems still lack standardized data collection and proper monitoring tools. The lack of accurate information about textile waste materials makes it difficult for stakeholders to identify opportunities for reuse or recycling[5].

1.2.2 Predictive Models for Waste Forecasting

Numerous researchers have investigated the application of machine learning and artificial intelligence to forecast trash creation trends and industrial production outcomes. In order to spot patterns and project future output or waste levels, predictive analytics programs can examine past data. The manufacturing sector is using these models more and more to enhance resource management and planning[6].

In the context of waste management, predictive models can help organizations anticipate future waste volumes and develop more effective recycling strategies. Machine learning approaches such as neural networks, regression models, and ensemble learning techniques have been applied to analyze industrial data and improve prediction accuracy[6].

Despite these advancements, most existing research focuses primarily on production optimization or energy consumption rather than specifically forecasting textile waste

quantities. This indicates a need for specialized predictive models designed for textile waste analysis.

1.2.3Analytical Reporting and Visualization

Data visualization and reporting tools are important components of modern data-driven systems. Analytical dashboards help organizations interpret complex datasets by presenting information through graphs, charts, and visual reports[7].

In waste management systems, visualization tools allow stakeholders to monitor waste generation patterns, evaluate recycling performance, and understand environmental impacts. By presenting predictive results and waste statistics in dashboards, organizations can quickly identify trends and make informed decisions regarding waste management strategies[7].

1.2.4.Data-Driven Decision Support for Circular Economy

Reducing waste and optimizing resource reuse through recycling and sustainable production methods are the main goals of the circular economy idea. By offering insights that aid in decision-making, researchers stress that digital technologies like artificial intelligence, data analytics, and digital platforms can enhance waste management[8].

To assist stakeholders in selecting the best methods for recycling and trash reduction, decision support systems incorporate data analytics, forecasting models, and reporting tools. These tools make it possible for businesses and decision-makers to assess waste patterns and create long-term solutions for managing textile waste[8].

1.3 Research Gap

1. **Lack of Comprehensive Waste Visibility:**
   Most existing systems and platforms focus on basic listing or disposal of textile waste but **do not provide detailed information about waste materials**, such as type, quantity, and availability. Without accurate and accessible data, stakeholders—such as manufacturers, recycling industries, and policymakers—cannot efficiently identify opportunities for reuse or redistribution. This **gap highlights the need for a system that ensures complete textile waste visibility**, which directly aligns with the first research objective.
2. **Insufficient Predictive Analytics for Waste Forecasting**
   While some studies have applied predictive models in manufacturing and general waste management, **very few specifically target textile waste forecasting**. Current methods primarily focus on production optimization or energy consumption rather than predicting future textile waste volumes[9]. This insufficiency creates challenges for planning reuse strategies, resource allocation, and sustainable recycling initiatives. The **proposed forecasting model** will fill this gap by analyzing historical textile waste data to identify trends and estimate future quantities.
3. **Limited Reporting and Visualization Capabilities**
   Many existing platforms provide minimal reporting features or basic textual summaries of waste data. There is a **lack of integrated dashboards and analytical reports** that can visually present waste patterns, trends, and predictions[10]. Without these tools, stakeholders struggle to interpret complex data and make informed decisions. This **report generation and visualization module** addresses this shortfall by providing interactive dashboards and automated analytical reports to enhance the understanding of textile waste data.
4. **Fragmented Decision Support Systems**
   Current digital solutions often operate independently and fail to combine waste visibility, forecasting, and reporting into a single platform. Consequently, stakeholders cannot leverage **data-driven insights to make effective decisions** for reuse, recycling, and circular economy strategies. The proposed system's **decision support feature** is designed to integrate these elements, providing actionable insights and enabling proactive planning and sustainable decision-making.

5. **Justification of Novelty and Creativity**

   The novelty of this research lies in its **integrated approach**, which combines:

6. Real-time textile waste visibility
7. Predictive forecasting of waste quantities
8. Analytical reporting and visualization
9. Data-driven decision support
10. No existing system or platform currently unifies all these aspects specifically for **textile waste management**, making this approach both creative and highly practical for promoting sustainability.
11. **Evidence-Based Argument**

    Literature shows that although digital and AI-based tools exist for general waste management, **textile-specific solutions are rare**. Studies indicate limited transparency, forecasting, and decision support in current platforms (MDPI, 2025; ScienceDirect, 2025). These findings support the need for a **comprehensive, integrated system** tailored to textile waste management.

2. Objectives

2 .1. Main Objective

To generate the management of textile waste by generating analytical reports and visual dashboards, and to use predictive analytics to support data-driven decision-making for sustainable textile waste reuse and recycling.

2.2. Specific Objectives

1. **Textile Waste Visibility –** To provide detailed information about textile waste materials, including type, quantity, and availability, to improve transparency in waste management.
2. **Waste Forecasting Model Development –** To develop a predictive forecasting model that analyses historical textile waste data to identify trends and estimate future textile waste quantities.
3. **Report Generation and Visualization –** To generate analytical reports and visual dashboards based on predicted textile waste data to help stakeholders understand waste patterns and potential reuse opportunities.
4. **Data-Driven Decision Support –** To support effective decision-making for textile waste reuse, recycling, and circular economy initiatives using analytics and insights derived from the system**.**

3. Methodology

The Marketplace Platform and Analytics module's methodology aims to create a digital platform for trading textile waste and a predictive waste-forecasting dashboard. In order to develop capabilities for displaying, purchasing, and selling textile waste, the platform first collects requirements from stakeholders, such as manufacturers, designers, and upcycles. The system architecture and user interface are then developed to offer a seamless, useful marketplace experience. Predictive analytics, which forecasts future trash production and reuse/recycling opportunities, is conducted in tandem with the collection and analysis of historical waste data. Stakeholders may make well-informed, data-driven decisions for sustainable textile waste management thanks to the dashboard's visual display of this data.

3.1 Data Collection and Sourcing

To support the objectives of generating reports, predictive analytics, and data-driven decision-making for textile waste management, data was collected directly from

- **Colleague-Provided Textile Waste Data:**
  Detailed information about textile waste materials, including type, quantity, quality, and availability, was obtained from colleagues working in manufacturing units, recycling operations, and small-to-medium enterprises (SMEs). This data supports **Textile Waste Visibility** and ensures that reports and analytics accurately reflect real-world waste streams.
- **Historical Waste Records & Academic References:**
  Verified records of past waste generation, reuse, and recycling trends were collected from colleagues' internal documentation and supplemented with academic literature. This dataset is used to **develop the Waste Forecasting Model**, enabling predictions of future waste volumes and planning for reuse or recycling strategies.
- **Guidelines and Industry Practices:**
  Information on sustainability benchmarks and waste management practices, including colleague-provided insights and publicly available frameworks (e.g., Higg Materials Index, Sri Lanka National Waste Policy, ISO 14000), ensures that **Data-Driven Decision Support** aligns with circular economy principles and supports informed decision-making for textile reuse and recycling.
- **Report Generation Support:**
  Colleague-provided data is structured and verified to allow automated **analytical report generation and visualization**, helping stakeholders interpret waste trends and make actionable decisions.

Table 1: Textile Waste Categories and Reuse/Recycling Trends

Material	Availability	Reuse Trend	Recycling Trend
Cotton	High	Increasing	Moderate
Polyester	Medium	Moderate	Stable
Nylon	Medium	Moderate	Moderate
Polyester Mix	Medium	Moderate	Low

Table 1 Textile Waste Categories and Reuse/Recycling Trends

3.2. Feature Engineering

- Feature engineering data means the process of selecting data and creating meaningful variables by getting data from sellers.
- Primary Features: These are directly collected from waste management platforms.

1. **Predicted Waste Volume**
    - **Definition:** Estimates the future amount of textile waste expected for each material.
    - **Supports:** Waste Forecasting, Report Generation, Decision Support
2. **Recyclable Waste Quantity**
    - **Definition:** Calculates the portion of total waste that can potentially be recycled.
    - **Supports:** Data-Driven Decision Support, Report Generation
3. **Reusable Waste Quantity**
    - **Definition:** Determines the portion of textile waste suitable for reuse in the marketplace.
    - **Supports:** Waste Forecasting, Decision Support, Reporting
4. **Remaining Waste Quantity**
    - **Definition:** Calculates waste that is neither reusable nor recyclable.
    - **Supports:** Reporting, Decision Support, Resource Planning
5. **Marketplace Material Availability**
    - **Definition**: Indicates how much waste is currently available for trade after sales.
    - **Supports:** Textile Waste Visibility, Report Generation, Decision Support

- **Derived Features / Target Variables:**
    - $\mathbf{Predicted_waste_volume} = \mathbf{Historical_waste_quantity} \times \mathbf{Waste_growth_rate}$

      Estimates the future amount of textile waste expected for each material based on past trends.
    - $\mathbf{Recyclable_waste_quantity} = \mathbf{Total_waste_quantity} \times (\mathbf{Recycling_rate} / \mathbf{100})$

      Calculates the amount of textile waste that can potentially be recycled.
    - $\mathbf{Reusable_waste_quantity} = \mathbf{Total_waste_quantity} \times (\mathbf{Reuse_rate} / \mathbf{100})$

      Determines the portion of textile waste suitable for reuse in the marketplace.
    - $\mathbf{Remaining_waste_quantity} = \mathbf{Total_waste_quantity} - (\mathbf{Recyclable_waste_quantity} + \mathbf{Reusable_waste_quantity})$
    - $\mathbf{Marketplace_material_availability} = \mathbf{Total_listed_waste_quantity} - \mathbf{Sold_waste_quantity}$

3.3. Model Development and Selection

Approach A: Statistical Trend Analysis (Baseline Method)This approach uses basic statistical and time-series analysis to identify historical trends in textile waste generation and marketplace listings.

Used For - Identifying historical textile waste generation patterns, understanding seasonal trends in textile waste availability, and providing a baseline comparison for forecasting models.

- **How It Works**
- **Input historical marketplace and waste data, such as:**
    - Waste quantity (kg)
    - Material type (cotton, polyester, blended)
- **Analyze patterns using statistical techniques such as:**
    - Moving averages
    - Trend analysis
    - Time-series visualization

    Example Formula

    Moving Average Forecast:

    Forecast(t+1) = $(Waste_1 + Waste_2 + \ldots + Waste_n) / n$

    - **Outputs**
    - Historical waste generation trends
    - Average textile waste availability
    - Seasonal waste patterns
    - Baseline waste forecast values

Approach B: Machine Learning Forecasting Model

This approach uses supervised machine learning regression models to predict future textile waste quantities and marketplace availability based on historical data.

Used For

- Predicting future textile waste generation
- Forecasting waste availability for buyers and recyclers

- Supporting data-driven decisions in the marketplace

How It Works

- **Input features such as:**
    - Historical waste quantities
    - Material types
    - Listing frequency
    - Location
    - Time (month/season)

Outputs

- Predicted textile waste quantities
- Future waste availability trends
- Forecast charts for the analytics dashboard

3.4. Model Training and Validation

Data Splitting

- The collected dataset related to **textile waste listings and historical waste quantities** will be divided into **training and testing datasets**.
- Typically, **80% of the data will be used for training** the model, while the **remaining 20% will be used for testing**.
- The **training dataset** is used to teach the model patterns in textile waste generation and marketplace listings.
- The **testing dataset** is used to evaluate how well the model can predict **future textile waste quantities and trends** on unseen data.

Training

- The following algorithms could be applied: •
- To find correlations between the amounts of textile waste and affecting factors, use linear regression.

    - Random Forest Regression: to identify more intricate trends in the production and accessibility of textile waste
- Gradient Boosting, also known as XGBoost, uses past waste trends to increase prediction accuracy.
- These models will be trained to **predict future textile waste quantities and marketplace availability**, and the prediction results will be displayed in the **waste forecasting dashboard to support data-driven decision-making for textile waste reuse and recycling**.

Validation

- Evaluate the trained models using the testing dataset.
- Measure performance using evaluation metrics such as R^2, MAE, and RMSE.

This helps check the accuracy of the model's predictions.

3.5. API Development and Integration

- **Endpoint:** A single POST endpoint, e.g., `GET /api/waste-forecast`
- **Request Body:** A JSON object containing the input features:

GET/api/waste-forecast?material_type=Polyester&location=
Colombo&start_date=2026-01-01&end_date=2026-02-28

- **Response Body:** A JSON object containing the predictions:

```
{
 "status": "success",
 "material_type": "Polyester",
 "location": "Colombo",
 "total_available_kg": 1200,
 "forecast_next_month_kg": 1350,
 "trend": "Increasing",
 "listings": [
  {
   "listing_id": "WL101",
   "quantity_kg": 500,
   "available_date": "2026-03-10"
  },
  {
   "listing_id": "WL102",
   "quantity_kg": 700,
   "available_date": "2026-03-12"
  }
 ]
}
```

- **Error Handling:** Error handling in API development ensures that the system can clearly manage unexpected issues such as invalid inputs, missing data, or server failures.
- **Documentation:** API documentation clearly describes how an API works, including available endpoints and error codes. It enables others to understand, integrate, and maintain the system efficiently, ensuring smooth communication between different software components.

4. High-level system architecture

4.1 Overview of Proposed Solution

The proposed solution is a **digital textile waste management system** designed to:

- Provide **detailed visibility** of textile waste materials (type, quantity, availability)
- Generate **predictive forecasts** of future waste volumes
- Produce **analytical reports and visual dashboards**
- Support **data-driven decision-making** for reuse, recycling, and circular economy initiatives

The system integrates **data collection, processing, analytics, and reporting modules** to create a cohesive workflow from raw data to actionable insights.

4.2 System Components

The system consists of the following **core components**:

1. **Data Collection Module**
    - **Function:** Collects primary data from colleagues including material type, quantity, quality, location, reuse/recycling potential, and historical waste records.
    - **Parts:**
        - Data Input Interface
        - Data Validation & Cleaning
2. **Waste Visibility Module**
    - **Function:** Displays real-time information about textile waste availability, type, and status.
    - **Parts:**
        - Material Catalog
        - Availability Tracker
        - Status Dashboard
3. **Predictive Analytics / Forecasting Module**
    - **Function:** Uses historical data to **predict future waste volumes** for each material type.
    - **Parts:**
        - Forecasting Engine (e.g., time-series or growth-rate models)
        - Calculated Primary Features (Predicted Waste, Reusable, Recyclable, Remaining Waste)
4. **Report Generation & Visualization Module**
    - **Function:** Produces analytical reports and dashboards to help stakeholders understand trends, reuse/recycling patterns, and decision insights.
    - **Parts:**
        - Automated Report Generator
        - Interactive Dashboards (Graphs, Charts, Tables)
5. **Decision Support Module**
    - **Function:** Provides actionable recommendations for stakeholders to reuse, recycle, or allocate waste.
    - **Parts:**

        - Insight Engine (based on predictions and reports)
        - Recommendations Interface
6. **Data Storage & Management Module**
    - **Function:** Stores all collected, processed, and calculated data securely.
    - **Parts:**
        - Relational Database (SQL/NoSQL)
        - Backup and Recovery Systems
        - Access Control

4.3 Component Interactions and Data Flow

1. **Data Collection Module → Waste Visibility Module**
    - Raw material data flows into the visibility module to display current availability and type.
2. **Data Collection Module → Predictive Analytics Module**
    - Historical and current waste data is processed to calculate **predicted waste volumes, reusable/recyclable quantities, and remaining waste**.
3. **Predictive Analytics Module → Report Generation & Visualization Module**
    - Forecasted data feeds into reports and dashboards to visualize trends, patterns, and material availability.
4. **Report Generation Module → Decision Support Module**
    - Insights and forecasts are analysed to provide **recommendations for reuse, recycling, and resource allocation**.
5. **Data Storage → All Modules**
    - Central database ensures all modules have access to **up-to-date and validated data**.

4.4 Data Flow Explanation

- **Input:** Raw textile waste data (type, quantity, quality, location, historical records)
- **Processing:** Cleaned, structured, and feature-engineered data → forecasting calculations → report generation
- **Output:**
    - Real-time **material availability information**
    - **Predicted waste trends**
    - **Analytical reports and dashboards**
    - **Actionable recommendations for decision-making**

4.5 Scalability and Security Considerations

- **Scalability:**
  - Modular design allows addition of new materials, locations, or users without major system changes.
  - Forecasting models and dashboards can be expanded to handle **large datasets** as more data is collected.
- **Security:**
  - Role-based access ensures **only authorized users** can view or modify sensitive data.
  - Database encryption and backup policies protect against data loss or leaks.
  - Data validation prevents inaccurate inputs from corrupting analytics.

4.6. System Overview Diagram

Figure 1: System Overview Diagram

Figure 1: The diagram shows the system flow of the textile waste marketplace platform. Manufacturers upload waste details to the platform, which are stored in a database. Buyers browse and purchase available materials. The analytics module processes historical data to analyze trends and predict future waste quantities, and the results are displayed in the forecasting dashboard for informed decision-making.

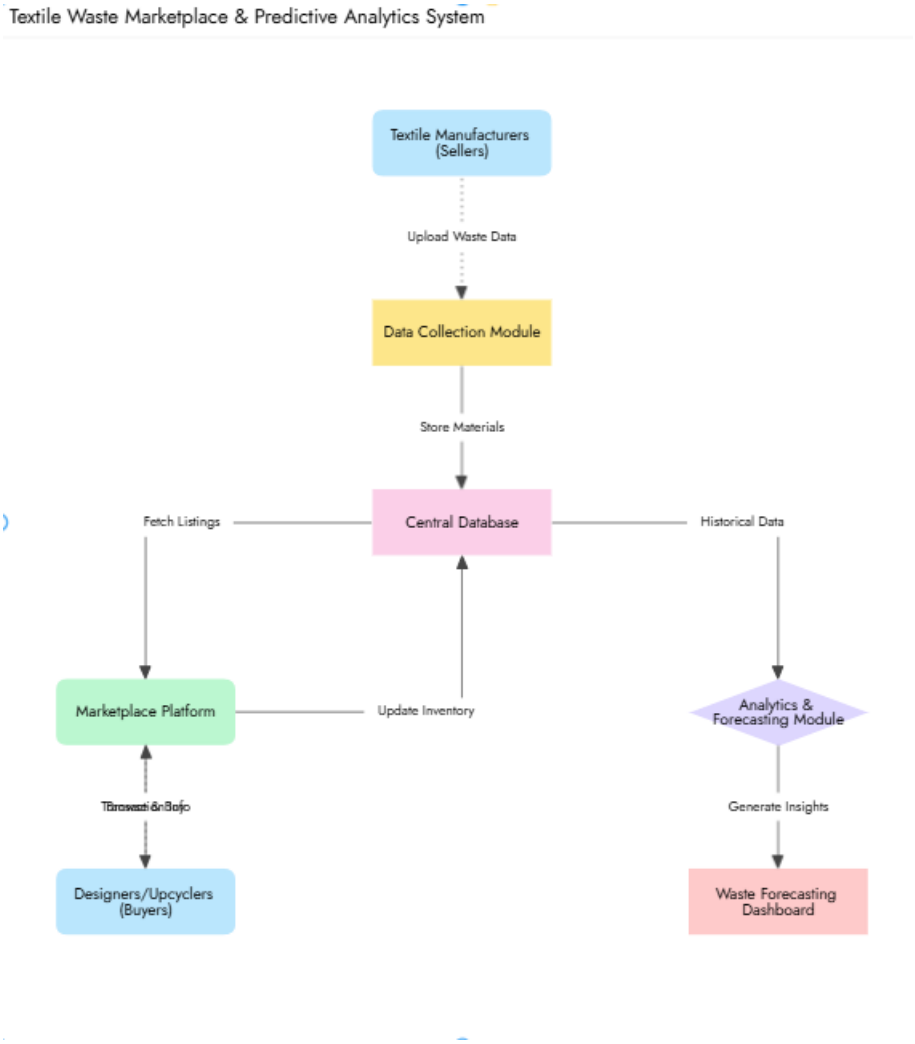


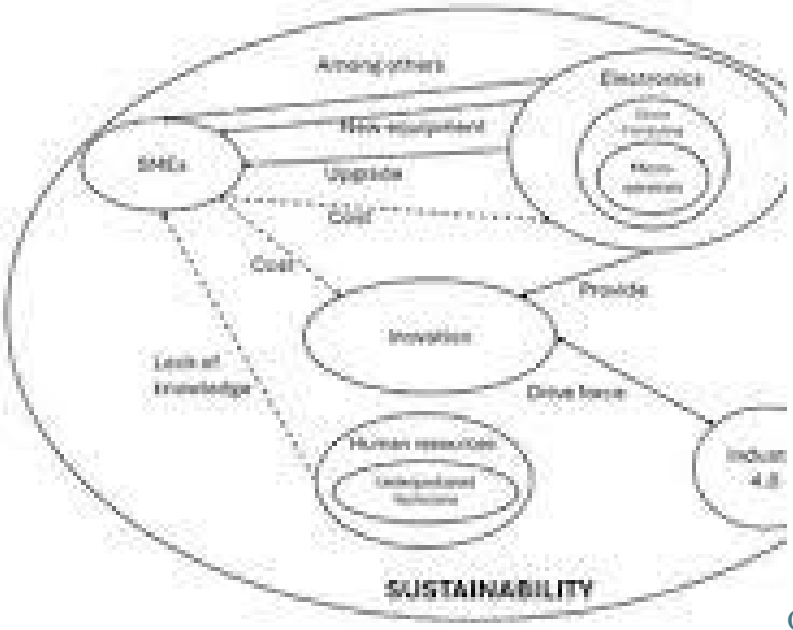
Figure 1 System Overview Diagram

5. Requirements

5.1. User Requirements

The platform's users are primarily buyers and sellers of textile waste.

- Buyers (designers, upcyclers, SMEs) need insights on textile waste availability, material types, and predicted future supply.
- Sellers (manufacturers, garment factories) need analytics to monitor waste trends and forecast volumes, helping them plan production and disposal efficiently.
- Both user groups require clear, visualized data and forecasts to support economically and environmentally informed decisions regarding textile reuse and recycling.



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5.2. Use Cases

Table 2: Use Case 1 – View Waste Forecast Dashboard

Element	**Description**
Use Case ID	UC-01
Name	View Waste Forecast Dashboard
Summary	Buyers, sellers, and stakeholders can view visualized trends and predicted waste quantities to support informed decision-making.
Priority	High
Preconditions	• User account is registered and logged in. • Historical waste data is available in the database. • Analytics module and dashboard are active. • System is online and accessible.
Postconditions	• Dashboard displays updated waste trends and forecasts. • Users can filter data by material type, location, or time period.

	• Forecasted waste quantities are available for download or export. • Insights support decision-making for buying, selling, and recycling.
Primary Actor(s)	Buyers, Sellers, Sustainability Stakeholders
Secondary Actor(s)	System, Analytics Module, Database
Trigger	User wants to analyze historical waste data and view predictions for future waste quantities.
Main Scenario	1. User logs in. 2. User navigates to the waste forecasting dashboard. 3. System retrieves historical waste data from the database. 4. Analytics module processes data and generates trends and forecasts. 5. Dashboard displays visual charts, tables, and key insights. 6. User applies filters as needed and downloads reports if required.System confirms listing creation.
Extensions	• Data unavailable → system shows "no data" message. • Forecasting module error → system displays error notification.
Open Issues	How to integrate predictive analytics for mixed-material waste forecasting.

Table 2 Use Case 1 – View Waste Forecast Dashboard

Table 3: Use Case 2 – Match Buyers to Waste Listings

Element	**Description**
Use Case ID	UC-02
Name	Match Buyers to Waste Listings
Summary	The system suggests potential buyers to factories based on textile waste type, quantity, and predicted availability.

Priority	High
Preconditions	• Factory account is registered and logged in. • Waste listing exists in the system. • Analytics and AI matching module are active. • Database is online and accessible.
Postconditions	• Potential buyers are suggested to the factory. • Matching recommendations are displayed on the platform. • The factory can contact or notify suggested buyers. • Matching records are stored for analytics.
Primary Actor(s)	Buyers, Recycler
Secondary Actor(s)	
Trigger	• The factory wants to find potential buyers for a newly submitted waste listing.
Main Scenario	1. Factory submits a new textile waste listing. 2. System validates and saves the listing in the database. 3. The AI matching module analyzes material type, quantity, and location. 4. System generates a list of potential buyers. 5. Recommendations are displayed to the factory. 6. Factory contacts or notifies selected buyers.
Extensions	• No suitable buyers → system notifies factory. • Buyer preferences change → system updates matching list.
Open Issues	How to rank suggested buyers based on location, demand, or environmental criteria.

Table 3 Use Case 2 – Match Buyers to Waste Listings

5.3. Wireframes

Figure2:Wireframe – Marketplace Listing Page

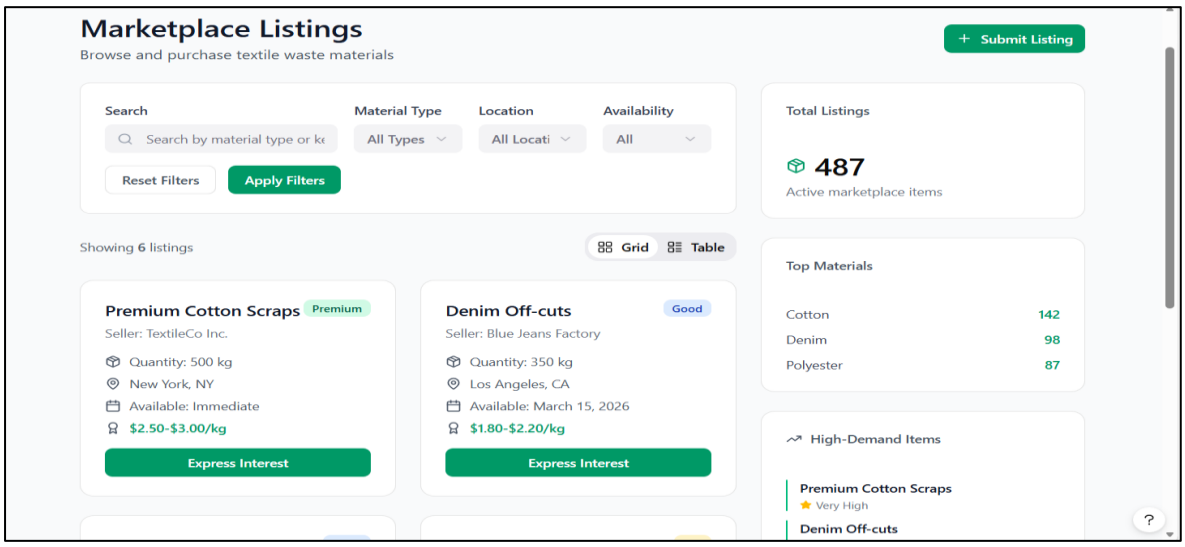


Figure 2 Wireframe – Marketplace Listing Page

Figure 3: Wireframe – Waste Forecasting Dashboard

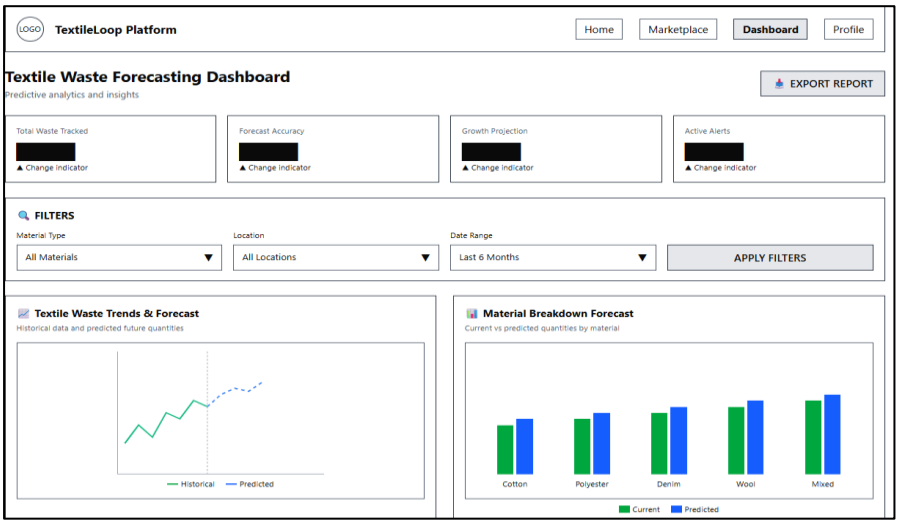


Figure 3 Wireframe – Waste Forecasting Dashboard

5.4. Functional Requirements

ID	Requirement Description
FR-01	The system shall allow textile waste data to be entered, including material type, quantity, location, and availability.
FR-02	The system shall store and manage all waste data in a central database.
FR-03	The platform shall provide a dashboard that visualizes textile waste trends and patterns.
FR-04	The system shall allow filtering and analysis by material type, quantity, and location.
FR-05	The platform shall analyze historical waste data to forecast future textile waste volumes for different materials.
FR-06	Predicted results shall be displayed on the dashboard in charts, graphs, and tables for easy interpretation.
FR-07	The system shall provide insights to support informed decision-making for buyers, sellers, and sustainability stakeholders.
FR-08	The dashboard shall highlight trends, predicted availability, and potential reuse/recycling volumes.

Table 4 Functional Requirements

5.5. Non-Functional Requirements

ID	Requirement Description
NFR-01	**Performance** • The platform shall process and display analytics and forecasts within 3–5 seconds for standard datasets. • The dashboard shall handle large datasets of textile waste without performance degradation.
NFR-02	**Usability** • The dashboard shall be intuitive and easy to use, with clear visualizations of waste trends and forecasts. • Users shall be able to filter and interpret data easily.
NFR-03	**Scalability** • The system shall support adding more materials, users, and datasets without major redesign. • The analytics module shall accommodate future predictive models or additional data sources.
NFR-04	**Reliability** • The platform shall ensure data integrity and accurate analytics even in the case of high traffic. • Forecasting and trend analysis shall produce consistent results when run multiple times with the same data.
NFR-05	**Security** • The system shall protect sensitive waste data from unauthorized access. • Data stored in the database shall be encrypted and backed up regularly.
NFR-06	**Maintainability** • The platform shall allow easy updates to analytics algorithms, dashboard features, and database schemas. • The system architecture shall support modular maintenance without downtime.
NFR-07	**Accessibility** • The dashboard shall be accessible via web browsers on desktop and mobile devices.

	• Visualizations shall follow standard accessibility guidelines for color, text, and readability.

Table 5 Non-Functional Requirements

6. Commercialization plan

Target Market:

- **Manufacturers & Textile Producers:** Companies generating large volumes of textile waste who want to sell or reuse materials efficiently.
- **Recyclers & SMEs:** Small-to-medium businesses focusing on textile recycling, upcycling, and resale of reusable materials.
- **Designers & Fashion Brands:** Businesses interested in sourcing sustainable materials for eco-friendly production.

Personnel: Factory Manager



Figure 4 Personnel: Factory Manager

Customer Persona:

- **Name:** Eco-conscious Textile Manager
- **Role:** Manages waste at a manufacturing facility or recycling business

- **Needs:** Transparent information on waste types, quantities, and availability

Persona 2: Eco-conscious Textile Manager



Figure 5 Persona 2: Eco-conscious Textile Manager

Value Proposition

- **Transparency:** Detailed visibility of textile waste types, quantities, and availability.
- **Predictive Insights:** Forecast future waste trends to plan for reuse or recycling.
- **Data-Driven Decisions:** Analytics and reports allow stakeholders to make informed, sustainable choices.
- **Efficiency:** Simplifies the management and trade of textile waste, reducing time and cost in sourcing or recycling.
- **Sustainability:** Supports circular economy initiatives, helping companies reduce environmental impact.

Revenue Model

- **Subscription-Based Model:** Monthly or annual subscription for access to dashboards, forecasts, and reports.
- **Pay-Per-Report:** Option to purchase individual analytical reports or forecast summaries.
- **Premium Features:** Advanced analytics, historical trend comparison, and decision support recommendations for higher-tier subscribers.
- **Consulting Services:** Advisory for waste reduction strategies and circular economy integration.

Cost Estimation

- **Development Costs:** Software development, data integration, analytics engine setup, and report generation system.
- **Operational Costs:** Data storage, server hosting, maintenance, and updates.
- **Marketing Costs:** Outreach to manufacturers, recyclers, and fashion brands.
- **Support Costs:** Customer support and training for users of the system.

Pricing Strategy

- **Tiered Subscription:**
    - Basic: Access to waste visibility and standard reports
    - Standard: Includes predictive analytics and interactive dashboards
    - Premium: Full decision support recommendations, advanced forecasting, and priority support
- **One-Time Report Purchase:** Affordable pricing for single-use analytical reports for smaller businesses.

Competitive Advantage

- **Integrated Solution:** Combines **waste visibility, forecasting, report generation, and decision support** in one system.
- **Data-Driven Insights:** Focuses on **actionable analytics**, not just raw data.
- **Sustainability Focus:** Directly supports **circular economy and reuse initiatives**, which most competitors do not provide.
- **Ease of Use:** Designed to simplify decision-making and reduce manual tracking or spreadsheet dependency.

Intellectual Property Considerations

- **Software & Algorithms:** Predictive models and report-generation algorithms can be protected under **copyright or trade secrets**.
- **Data Models:** Proprietary data processing methods for textile waste analytics can be **internally documented for IP protection**.
- **Branding & Trademarks:** Company name, logo, and dashboard interface can be **trademarked** to protect the system's identity.
- **Licensing:** Consider licensing the software or predictive analytics models to third-party businesses for additional revenue.

7. Budget and justification

Component	Amount (USD)	Amount (LKR)	Justification
Data and Software			
Textile Waste Dataset Sources (Reports & Public Data))	0.00	0.00	Data related to textile waste quantities, material types, and availability will be collected from publicly available research papers, sustainability reports, and open datasets.
Data Analysis Tools	0.00	0.00	Data processing and analytics will be performed using open-source tools such as Python, Pandas, and data visualization libraries.
Infrastructure and Tools			
Cloud Computing (AWS / Google Cloud / Render)	25.00	7,500.00	Cloud services (AWS, Google Cloud, or Render) will be used to host the marketplace prototype and analytics dashboard. Free tiers will support most development and testing activities.
Development Tools (IDE, Git, Frameworks)	0.00	0.00	Development will use free tools such as VS Code, GitHub, JavaScript frameworks, and open-source libraries for building the platform and dashboard.
Internet Charges (Research & Development)	15.00	4,500.00	Internet connectivity is required for literature review, dataset collection, cloud deployment, and collaboration during the research period.
Knowledge and Consultation			
Data Analytics / Dashboard Design Guidance	15.00	4500.00	Consultation with an AI/ML expert for model selection, feature engineering, and evaluation methods.
Textile Industry Insights	10.00	3000.00	Consultation with sustainability or textile experts to validate environmental impact

			assumptions and circular economy practices.
Contingency			
Miscellaneous Expenses	10.00	3,000.00	Covers unexpected costs such as additional cloud usage, data processing resources, or documentation printing.
TOTAL	**75.00**	**22,500.00**	The estimated total budget required for the research and prototype development.

Table 6 Budget and justification

8. Work breakdown structure (WBS)

	Task Description	**Subtask Description**
1.0	Textile Waste Management System	Main system to manage and analyze textile waste for visibility, forecasting, reporting, and decision support.
1.1	Research & Planning	Initial research and requirement analysis phase for the system
1.1.1	Review Existing Textile Waste Management Studies	Analyze current methods, platforms, and gaps in textile waste management
1.1.2	Define Objectives & System Requirements	Establish project objectives and functional requirements aligned with research goals.
1.2	Data Collection & Preparation	Collecting from team members and preparing data for the system and analytics modules.
1.2.1	Collect Textile Waste Data from Team Members	Gather data on type, quantity, and availability from team members, manufacturers, and recyclers.
1.2.2	Integrate Data into Marketplace Platform	Input the processed data into the digital marketplace for visibility and trade.
1.3	Waste Forecasting Model Development	Build and validate predictive models for future textile waste quantities.

1.3.1	Analyze Historical Textile Waste Data	Explore and extract trends from historical datasets.
1.3.2	Develop Predictive Forecasting Model	Implement statistical or machine learning models to forecast waste.
1.3.3	Validate & Test Model	Assess model accuracy and reliability using testing datasets.
1.4	Report Generation & Visualization	Create visual dashboards and automated reports for stakeholders.
1.4.1	Design Dashboards & Reports	Develop templates and visualization layouts for analytics outputs.
1.4.2	Generate Analytical Reports	Produce reports based on predictive model outputs and collected data.
1.4.3	Integrate Visualization with Marketplace	Connect dashboards to the platform for interactive stakeholder access.
1.5	Data-Driven Decision Support	Support decision-making for textile reuse, recycling, and circular economy initiatives.
1.5.1	Develop Decision Support Tools	Build analytics modules to provide actionable insights.

1.5.2	Implement Recommendations	Generate data-driven strategies for stakeholders based on analytics.
1.5.3	Final System Integration & Testing	Ensure all components (visibility, forecasting, reporting, decision support) work seamlessly together.
1.6	Deployment & Documentation	Prepare the system for use and create comprehensive documentation.
1.6.1	Deploy System & Marketplace Platform	Make the system accessible to users and stakeholders.
1.6.2	Write Methodology & Results Documentation	Document the methodology, analysis, and findings of the project.
1.6.3	Prepare Final Presentation & Demonstration	Create presentation materials and a demo for evaluation or submission.

Table 7 Work breakdown structure (WBS)

9. Gantt chart

Objective	Task Description	Start Week	End Week	Duration (Weeks)
1. Textile Waste Visibility	Collect data from manufacturers, recyclers, and SMEs	1	3	3
	Organize and preprocess data	3	5	3
	Integrate data into the marketplace platform	5	7	3
2. Waste Forecasting Model Development	Analyze historical textile waste data	5	7	3
	Develop a predictive forecasting model	7	10	4
	Validate and test model predictions	10	12	3
3. Report Generation and Visualization	Design dashboards and report templates	8	11	4
	Generate analytical reports from model output	11	13	3
	Integrate visualization with the marketplace platform	12	14	3
4. Data-Driven Decision Support	Develop decision-support analytics tools	13	16	4
	Implement recommendations based on analytics	15	18	4
	Final integration & testing	17	20	4

Table 8 Gantt Chart

Figure 4: Gantt Chart

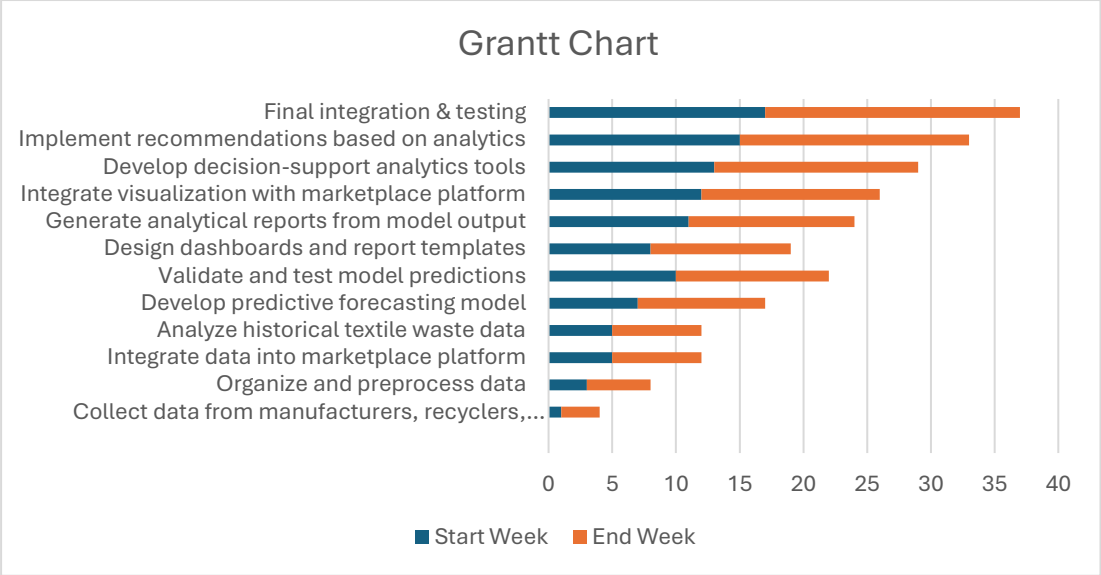


Figure 6 Gantt Chart

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